

Design Proposal

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I. INTRODUCTION

Included in my draft layout are the following devices:

a) 1550 nm MZI comprising two 50/50 directional couplers with a 300 um path length difference. For this device, all components are provided in the ebeam PDK for layout and simulation.

b) Add-drop ring resonator $R = 18 \text{ um}$, $g = 200 \text{ nm}$. This component is provided in the ebeam PDK and the S parameters are included for simulation in interconnect.

c) Add-drop ring resonator $R = 10 \text{ um}$, $g = 200 \text{ nm}$. This component is provided in the ebeam PDK but the S parameters are not included for interconnect simulation. These are simulated using FDTD and then imported into Interconnect for simulation.

Pending time and space I will add more devices to my final layout.

II. THEORY

The transfer function for the two ports of an ideal lossless MZI are given by the following equations:

$$P_1 = \frac{1}{2}(1 + \cos 2\theta) \quad (1)$$

$$P_2 = \frac{1}{2}(1 - \cos 2\theta) \quad (2)$$

Where P_1 is the power measured on output port 1 and P_2 is the power measured on output port 2. $\theta = \frac{\Delta L \pi n_{eff}}{\lambda}$ is the phase difference between the two arms of the MZI as a function of the path length difference between the arms, ΔL and the wavelength, λ . The free spectral range of the MZI is given by:

$$\Delta\lambda = \frac{\lambda^2}{\Delta L n_g}, \quad (3)$$

where n_g is the group index of the waveguide.

A ring resonator can be used as an add-drop filter allowing light to pass through to the drop port only on resonance. Resonances are seen as dips in the spectrum of the through port and peaks in the spectra of the drop port and occur at wavelengths separated by the FSR of the ring resonator.

The transfer function for the through port of the ring resonator is given by:

$$P_T = \quad (4)$$

$$P_D = \quad (5)$$

And the free spectral range of the ring resonator is given by:

$$\Delta\lambda = \frac{\lambda^2}{2\pi R n_g}, \quad (6)$$

where R is the ring radius and n_g is the group index.

III. MODELLING AND SIMULATION

The waveguides used for these devices are silicon-on-insulator strip waveguides measuring 220 nm thick and 500 nm wide. The geometry of these waveguides is shown in figure 1, and the TE mode that propagates in them at 1550 nm as simulated in Lumerical MODE is depicted in figure 2. All devices described in this design proposal have been designed for the TE mode.

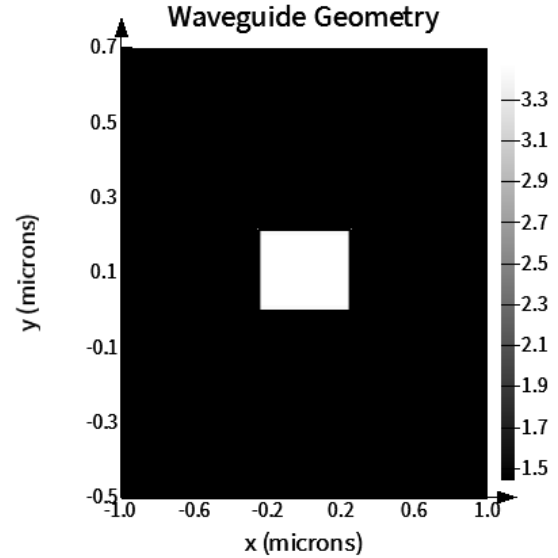


Fig. 1. The geometry of the 1550 nm waveguides used for the devices described here.

The effective index of these waveguides as a function of wavelength is shown in figure 3 and the group index as simulated in Lumerical MODE is shown in figure 4.

The wavelength dependence of the effective index can be approximated by fitting a Taylor series expansion to the data resulting in a compact model. Using matlab the compact model for the waveguides described here is found to be:

$$n_{eff}(\lambda) = 2.447 - 1.133 * (\lambda - 1.55) - 0.444\lambda - 1.55)^2$$

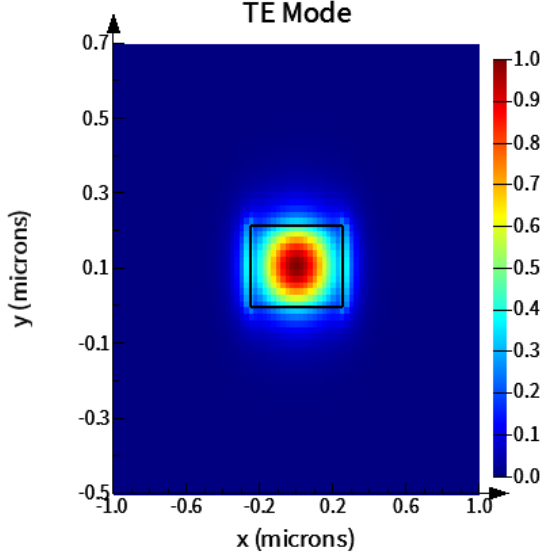


Fig. 2. A simulation of the TE mode propagating inside of a straight waveguide at 1550 nm. Simulated using Lumerical MODE.

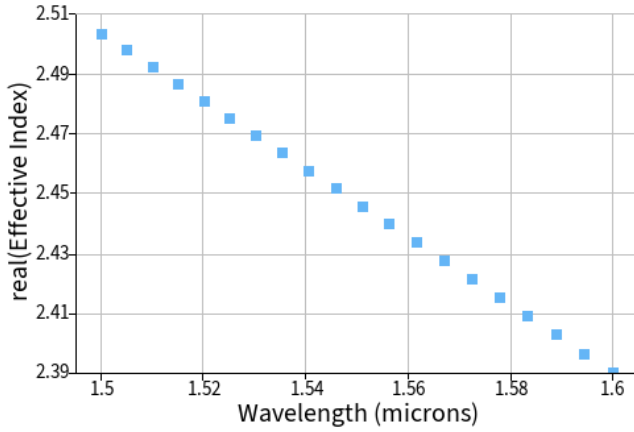


Fig. 3. The effective index of the waveguides used for these devices as a function of wavelength simulated using Lumerical MODE.

The first device is an unbalanced MZI comprising of two directional couplers designed for 1550 nm wavelength (ebeam_dc_te1550), light is coupled into the device using grating couplers designed for 1550 nm to polarized light (ebeam_gc_te1550). The short arm of this device is 300 μm long and the long arm is 600 μm , resulting in $\Delta L = 300 \mu\text{m}$. This gives us a theoretical FSR of 1.91 nm, as calculated from equation 3. The results of a Lumerical Interconnect simulation of this device is depicted in figures 6 and 7. The diagram depicting the simulation of this device is shown in figure 5. The simulations show an FSR of 1.88 nm.

The second device is an add-drop ring resonator with a radius of $R = 10 \mu\text{m}$ and a coupling gap of $g = 200 \text{ nm}$ it is made using two half-rings connected together to form a full ring (ebeam_dc_halfiring_straight_1), light is coupled into the device using grating couplers designed for 1550 nm te

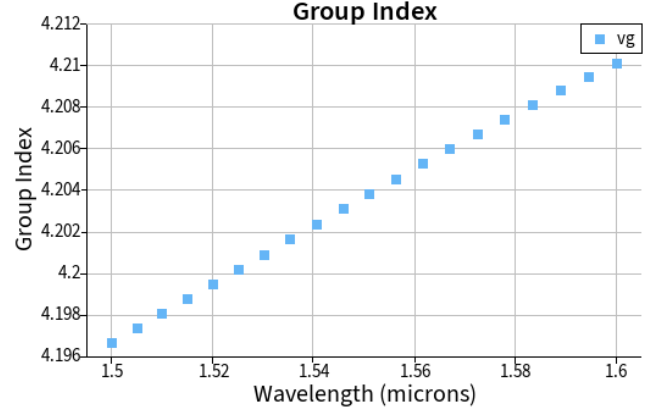


Fig. 4. The group index of the waveguides used for these devices as a function of wavelength simulated using Lumerical MODE.

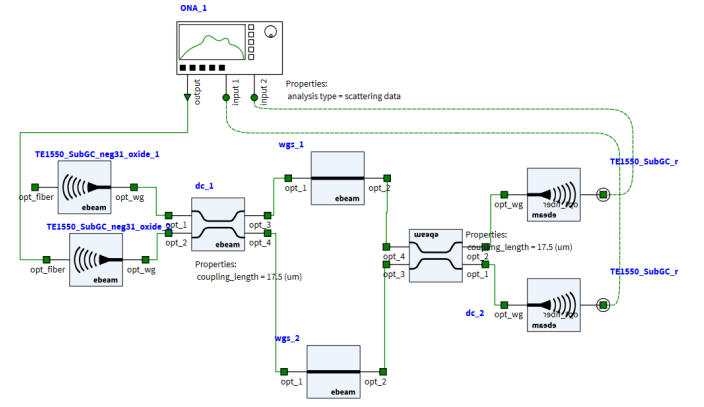


Fig. 5. Diagram of the setup for the simulation of the MZI with $\Delta L = 300 \mu\text{m}$ in Lumerical Interconnect.

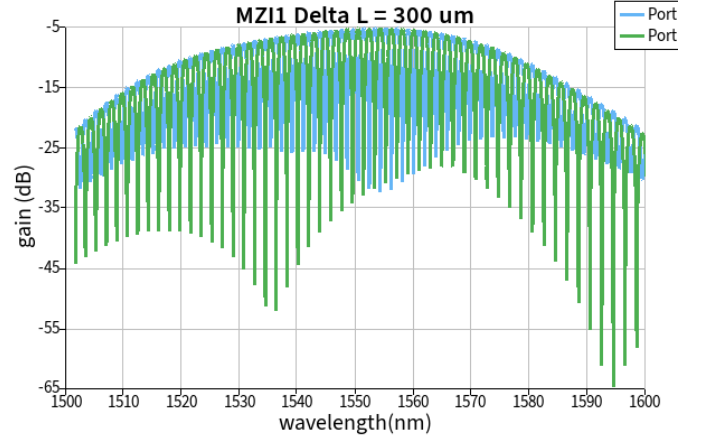


Fig. 6. The gain as a function of wavelength as simulated in Lumerical Interconnect over the range 1500 nm - 1600 nm.

polarized light (ebeam_gc_te1550). The theoretical FSR of this device is 9.10 nm, as calculated from equation 6. The S parameters for this device were not included with the ebeam package and therefore had to be simulated using Lumerical FDTD. This was done using the script ebeam_dc_halfiring_straight.lsf,

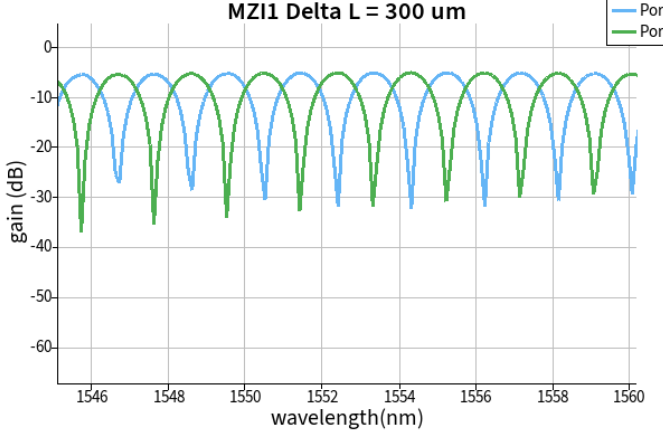


Fig. 7. The gain as a function of wavelength as simulated in Lumerical Interconnect zoomed into the range around 1550 nm.

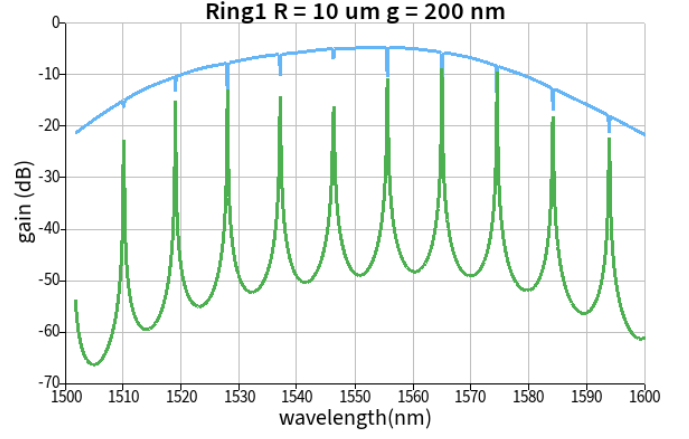


Fig. 9. Gain as a function of wavelength as simulated in Lumerical Interconnect over the range 1500 nm - 1600 nm. The Through port gain is shown in blue and the drop port gain is shown in green.

downloaded from the ebeam PDK repository. The correct parameters for this ring were input to the script, and it was then run with a mesh accuracy of 3. The resulting S parameters were saved and loaded into an Optical N Port S parameter component in Lumerical Interconnect. A diagram showing the setup for this simulation is given in figure 8, and the results of the Lumerical Interconnect simulation for this device are shown in figure 9. The simulations give an FSR of 9.65 nm.

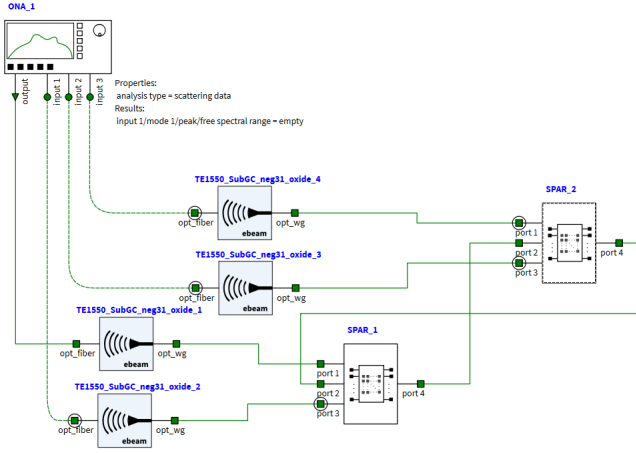


Fig. 8. Diagram of the setup for the simulation of the ring resonator with $R = 10\mu\text{m}$, $g = 200\text{nm}$ in Lumerical Interconnect.

The third device is an add-drop ring resonator with a radius of $R = 18\mu\text{m}$ and a coupling gap of $g = 200\text{nm}$ it is made using two half-rings connected together to form a full ring (ebeam_dc_halfing_straight_1), light is coupled into the device using grating couplers designed for 1550 nm te polarized light (ebeam_gc_te1550). The theoretical FSR of this device is 5.06 nm, as calculated from equation 6. The S parameters for this device were included with the ebeam package and so the device could be directly simulated in Lumerical Interconnect using the ebeam_dc_halfing_straight component. A diagram showing the setup for this simulation is given in figure 10, and the results of the Lumerical Interconnect simulation for this

device are shown in figure 11. The simulations give an FSR of 5.36 nm.

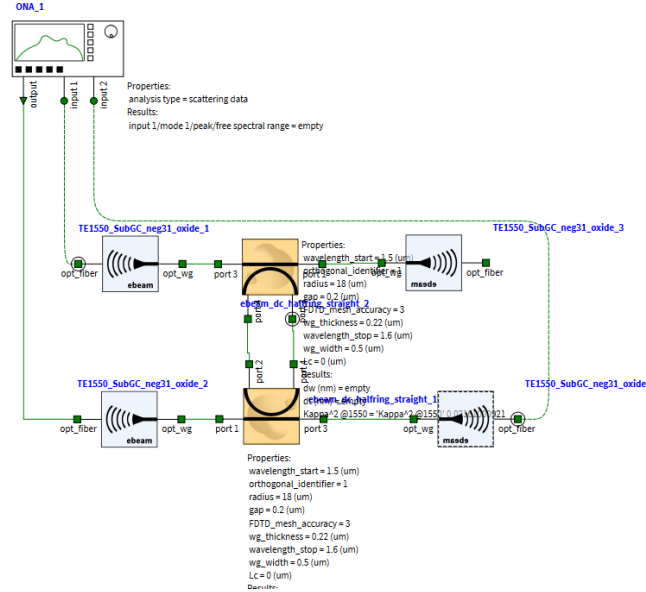


Fig. 10. Diagram of the setup for the simulation of the ring resonator with $R = 18\mu\text{m}$, $g = 200\text{nm}$ in Lumerical Interconnect.

The table below summarizes the theoretical and simulated free spectral ranges for the devices described and simulated above.

TABLE I
DEVICE PARAMETERS AND EXPECTED FREE SPECTRAL RANGE

Device	Parameters	Theoretical FSR	Simulated FSR
MZI1	$\Delta L = 300\mu\text{m}$	1.91 nm	1.88 nm
Ring1	$R = 10\mu\text{m}$, gap 200 nm	9.10 nm	9.65 nm
Ring2	$R = 18\mu\text{m}$, gap 200 nm	5.06 nm	5.36 nm

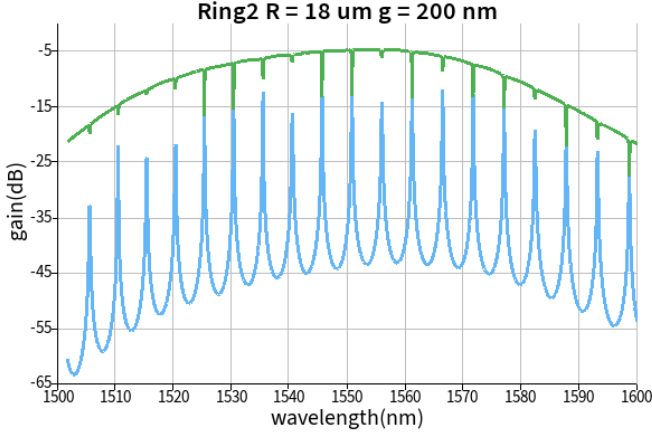


Fig. 11. Gain as a function of wavelength as simulated in Lumerical Interconnect over the range 1500 nm - 1600 nm. The Through port gain is shown in green and the drop port gain is shown in blue.

IV. FABRICATION

V. EXPERIMENTAL DATA

VI. ANALYSIS

Once we have the data we will be able to measure the FSR of each device by looking at the wavelength of two successive transmission peaks as close to the design wavelength of 1550 nm as possible. Once we have an experimentally measured FSR the experimentally measured group index of the MZI can be found by rearranging equation 3 to give:

$$n_g = \frac{\lambda^2}{\Delta L \Delta \lambda}, \quad (7)$$

and for the ring resonators by rearranging equation 6 to give:

$$n_g = \frac{\lambda^2}{2\pi R \Delta \lambda} \quad (8)$$

Since we don't have experimentally measured values yet, we can instead use our simulated values to obtain $n_g = 4.26, 3.96, 3.96$ for the first, second, third device respectively.

VII. CONCLUSION